

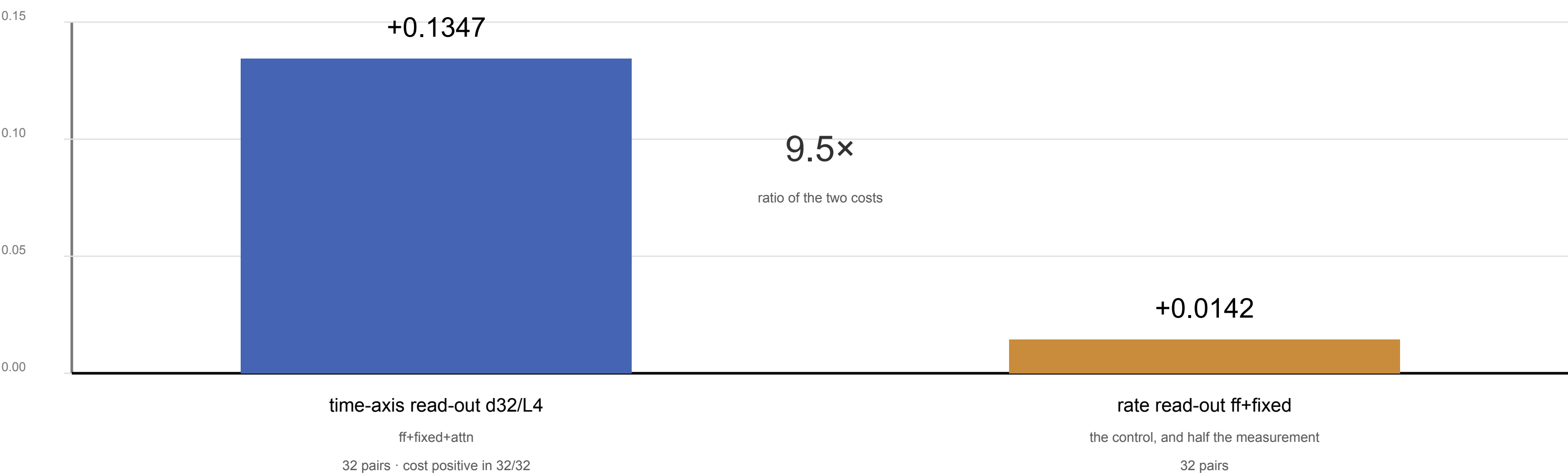
Figure 1 — The conditional: a difference-in-differences on the gain

Destroying temporal order costs the time-axis read-out almost everything it was buying, and costs the rate read-out almost nothing. The plotted quantity is each arm's own shuffle cost.

NOT SHOWN HERE, AND NOT CLAIMED: that SHD depends on temporal order. That is established — Cramer et al. 2022 (≤60% on spike-count-only SHD), the Neuromorphic Sequential Arena 2025 (86.48 → 68.51 with temporal processing removed model-side), Yu et al. 2025 (randomised spike times, reversed time).

Panel A — each arm's own shuffle cost. n = 32, seed-paired, h128 / published-2ms / adjacent-sum-5 / e400.

Bin-shuffling permutes time bins independently per sample in BOTH the training and test splits, so the task itself becomes rate-solvable. Nothing is removed from the model.



Panel B — the same result read as the collapse of the advantage over the rate arm.

n = 32	intact +0.1275	→	bin-shuffled +0.0070	94.5% of the advantage is contingent on temporal order
n = 12 (the registration this confirms)	intact +0.1258	→	bin-shuffled +0.0050	96% of the advantage is contingent on temporal order

Twenty further seeds move the gain by +0.0017 and the shuffle cost by +0.0010. The larger run confirms the registration; it did not rescue it.

Panel C — the n = 12 measurement this confirms, in absolute accuracy. LABELLED n = 12: absolute bin-shuffled means are not published at n = 32.

time-axis read-out 0.8320 → 0.6983 cost +0.1337, positive in 12/12

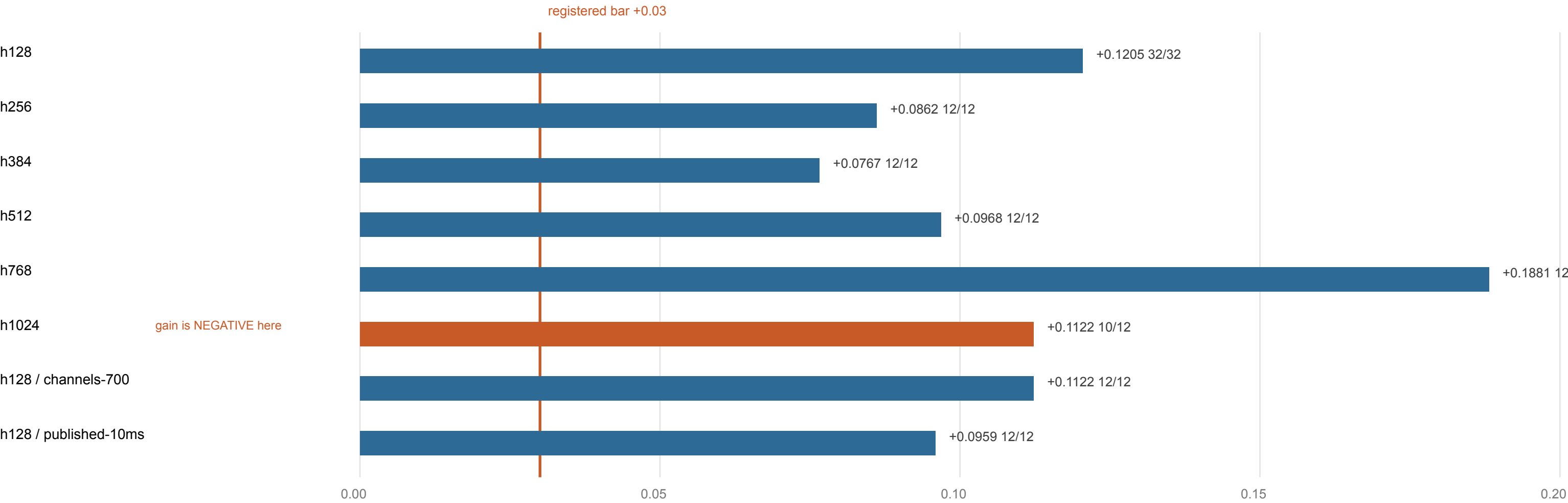
rate read-out 0.7062 → 0.6934 cost +0.0128

Per seed the effect falls between +0.0967 and +0.1568 — no seed in which it is absent. Every w9shf cell passes the temporal audit (counts preserved, relocated fraction ≥ 0.5).

What is measured here is which component's contribution is the order-dependent one: a difference-in-differences on the gain, not on accuracy.

Panel D — the same contrast at eight operating points. n = 12 per point, n = 32 at the anchor, seed-paired.

Only the bin-shuffled halves are new; intact halves are reused from the corpus at the same seeds and the same pinned binary.



The mechanism travels: every point clears the registered bar at BOTH budgets, and coverage is 21 of 21 operating points — NOT every width and NOT every read-out depth: h768/d32l2 is 0.1271 from its d32l4 twin, outside the 0.10 bar.

Its SIZE is not the gain. Spearman rho over the six widths is -0.1430 against a registered bar of +0.829 (NOT MET): h768 carries the smallest positive gain on the ladder (+0.0560) and the largest DiD here (+0.1881).

Points are in ladder order and are deliberately NOT connected and NOT sorted by effect size: either would assert a relationship the rank correlation rejects.